# EstateIQ-DIF: An Adaptive Multi-Modal Telemetry and Robust Statistical Decision Framework for Real-Time Commercial Building Anomaly Triage

*[Author Name(s) Placeholder] • Department of Electrical and Computer Engineering / Smart Facilities Lab • [Author Affiliation / Institution Placeholder] • Contact: [Author Contact / Email Placeholder]*
*Target Venue: [Target IEEE Conference Placeholder - Generic Research Draft] • Funding: [Funding Sponsor / Grant Details Placeholder - None Declared]*

***Abstract*—**This paper presents the EstateIQ Decision Intelligence Framework (EstateIQ-DIF), an interpretable, edge-compatible methodology for high-frequency contextual baseline modeling, operational rule integration, adaptive telemetry reporting, and anomaly triage across commercial facilities. Smart campuses scaling toward automated fault detection and diagnostics (FDD) face (1) a conflict between high-frequency telemetry bandwidth/storage costs and real-time fault-escalation latency, and (2) a tension between simplistic static thresholds that cause excessive false alarms and opaque deep-learning models that cannot be audited under facility compliance mandates. EstateIQ-DIF resolves both through a deterministic multi-stage pipeline: (1) building-specific seasonal baselines via group medians across 672 weekly quarter-hour slots; (2) robust dispersion from the Median Absolute Deviation (MAD) scaled by 1.4826, with dual relative and absolute scale floors; (3) deterministic cold-start fallback heuristics using occupancy and temperature adjustments; (4) bounded, two-sided exponential residual scoring; (5) domain operating-rule evidence signals; (6) convex score fusion; (7) bounded temporal persistence tracking with least-recently-used cache eviction; (8) an Adaptive Telemetry Cadence State Machine toggling between 15-minute nominal power-saving reporting (900 s) and 5-minute high-resolution anomaly monitoring (300 s) with a 3-cycle hysteresis recovery; and (9) an engineered physical FreeRTOS edge gateway integrating Modbus RTU electrical submetering, ultrasonic fluid-level sonar, turbine flow meters, and impedance-matched environmental sensing. On 350,400 simulated 15-minute electricity readings across ten buildings, a 7-day seasonal-naive baseline achieves MAE 1.5910 kWh per interval ($R^2$ = 0.9875), outperforming 15-minute persistence (2.1840 kWh) and the split mean (9.8093 kWh). Across injected fault episodes (426 intervals, 42 positive anomaly points), a pre-specified score threshold of 55/100 achieves precision 0.7069, recall 0.9762, F1-score 0.8200, false positive rate 0.0443, and ROC-AUC 0.9846, whereas unsegmented 7-day rolling Z-score and static historical P99 thresholds fail completely (F1 = 0.0000). In a sequential time-ordered holdout replay with transient motor inrush spikes, the adaptive engine achieves 100.0% specificity (0 false alarms) versus 96.9% for static thresholds. Adaptive cadence switching yields a 66.7% reduction in network transmission volume and radio wake cycles during nominal operation while guaranteeing sub-second local edge fault escalation.

***Index Terms*—**Building energy management, anomaly detection, robust statistics, median absolute deviation, seasonal baseline, telemetry processing, fault detection and diagnostics (FDD), decision support. *Draft Notice: generic IEEE conference-style draft; venue template not yet selected; results evaluated on generated telemetry.*

## I. INTRODUCTION

Modern commercial facilities and university campuses deploy advanced metering infrastructure (AMI) and multi-modal IoT sensor arrays capturing power, water and environmental metrics at sub-hourly intervals, which operators use to maintain efficiency, identify equipment malfunction, mitigate peak-demand charges and support decarbonization. Two persistent challenges arise:

- **Telemetry Transmission and Ingestion Bottlenecks:** with hundreds of electrical submeters, water flow turbines and environmental sensors, continuous 1- or 5-minute transmission exhausts wireless bandwidth, drains edge transceiver reserves and inflates cloud ingestion/storage costs; fixed 15-minute or hourly cadences create dangerous blind spots during pipe ruptures, phase overloads or pump failures.
- **Analytical Fragility of Static vs. Black-Box Detectors:** static or time-of-day BMS power thresholds cannot accommodate campus heterogeneity or climatic variation and generate alarm floods and operator fatigue; black-box models (deep recurrent networks, temporal transformers, autoencoders) need labeled fault datasets (exceptionally rare in commercial real estate), are vulnerable to sensor calibration drift, demand GPU acceleration, and offer limited interpretability when justifying a costly maintenance dispatch.

EstateIQ-DIF formulates contextual anomaly triage as a transparent, auditable, computationally efficient pipeline coupled with an adaptive edge-cloud telemetry reporting protocol. DIF operates without gradient-boosted trees (XGBoost, CatBoost, LightGBM), Prophet ensembles, Isolation Forests, LOF models, or SHAP attributions in its real-time scoring path. To prevent overclaiming, Table I classifies capabilities into three validation tiers: (i) evaluated in this paper; (ii) implemented in the repository but not evaluated here; (iii) demo, illustrative, or roadmap claims.

**TABLE I**
**PLATFORM SCOPE AND SCIENTIFIC VALIDATION BOUNDARIES**

| Feature / Subsystem — Presentation Role | Paper Validation Tier | Concrete Implementation & Scope |
|---|---|---|
| **Electricity Anomaly Triage** — real-time 15-min surge triage | Evaluated in this Paper | 672-slot median-MAD baseline, residual scoring, rule fusion, persistence gating |
| **Seasonal Baseline Modeling** — 7-day periodic profile estimation | Evaluated in this Paper | 15-min weekly slots with robust scale floors (ρ = 0.05, $\sigma_{abs}$ = 0.5 kWh) |
| **Adaptive Telemetry Cadence** — 15m nominal ↔ 5m anomaly | Evaluated in this Paper | State machine ($\Theta_t$), hysteresis recovery (M=3), 66.7% packet reduction |
| **Edge Hardware Gateway** — ESP32 FreeRTOS edge metrology | Evaluated in this Paper | Dual-core ESP32, Modbus RTU RS-485, ultrasonic sonar, Hall flow, ADC divider |
| **Multi-Scale Spatial Limits** — submeter vs campus feeder | Evaluated in this Paper | 3.5 kW submeter capacity vs 120.0 kW campus main; energy/power separation |
| **Energy Load Forecasting** — multi-step predictive load modeling | Evaluated in this Paper | 7-day seasonal naive benchmark across 50,304 holdout intervals ($R^2$ = 0.9875) |
| **Multi-Building Scalability** — campus-wide deployment | Evaluated in this Paper | 8 held-out buildings across 56 building-days (0.04% persistence FPR) |
| **Monotonic Counter Resilience** — hardware reboot & rollover | Evaluated in this Paper | Delta rollover isolation, microsecond timestamp monotonicity verification |
| **Water Consumption Triage** — flow spike & leak triage | Implemented; Unevaluated | Rule heuristics in `src/intelligence/water_engine.py`; not evaluated here |
| **Waste Fill-Level Monitoring** — bin overflow prediction | Implemented; Unevaluated | Heuristic checks in `waste_readings.csv`; not evaluated here |
| **Tree-Based SHAP Explainability** — offline root-cause attribution | Implemented; Unevaluated | TreeExplainer in offline diagnostic tools; excluded from real-time scoring path |
| **What-If Constraint Simulator** — HVAC setback scenario analysis | Implemented; Unevaluated | Deterministic rule simulation in `src/intelligence/scenarios.py`; not evaluated here |
| **Recommendation Ranking** — automated action utility scoring | Implemented; Unevaluated | Multi-criteria scoring in `src/intelligence/recommendations.py`; unevaluated |
| **Closed-Loop Verification** — post-action savings verification | Implemented; Unevaluated | Pre/post change-point verification engine; not evaluated on empirical telemetry |
| **Groq LLM Facility Copilot** — conversational natural-language UI | Demo / Illustrative Claim | UI presentation feature; not part of core mathematical scoring pipeline |
| **Commercial Deployment Mode** — cloud SaaS vs. on-premises licensing | Commercial Roadmap Claim | Business model proposal; not an empirical research claim |
| **Financial Savings Realization** — ₹9.50/kWh cost reduction impact | Hypothetical Scenario Claim | Continuation scenario arithmetic; does not reflect causal measured savings |

**Primary contributions:**

- **Code-Grounded Methodology:** rigorous specification of robust seasonal median-MAD baselining, dual scale flooring, deterministic cold-start estimation, bounded residual scoring, domain operating rules, weighted evidence fusion, bounded temporal persistence.
- **Adaptive Telemetry Reporting Protocol:** cadence state machine toggling 15-minute nominal and 5-minute anomaly/recovery reporting with 3-cycle hysteresis, giving a 66.7% payload reduction with sub-second edge alert latency.
- **Physical Edge Gateway Architecture:** ESP32 dual-core FreeRTOS gateway interfacing Modbus RTU power meters, ultrasonic tank transducers, turbine flow meters and impedance-matched environmental sensors.
- **Leakage-Conscious Evaluation Protocol:** chronologically segmented benchmark on simulated multi-building telemetry with documented provenance, verified file hashes and explicit hyperparameter disclosure.
- **Multi-Scale Spatial Physical Limits:** formal separation of submeter overcurrent limits (3.5 kW) from campus feeder limits (120.0 kW) and of instantaneous power (kW) from cumulative energy (kWh).
- **Systematic Ablations and Baseline Comparisons:** EstateIQ-DIF vs. rolling Z-score, static percentile and parametric Gaussian baselines, with ablations of residual scoring, operating rules and persistence gating on identical windows.
- **Sequential Holdout Replay Benchmark:** contextual MAD with temporal persistence fully rejects transient motor inrush spikes (100.0% specificity) whereas naive static thresholds raise false alarms.
- **Complexity and Microbenchmark Analysis:** expected O(1) time/memory per event; CPU latencies under 2 ms; peak heap 4.52 MB.

## II. RELATED WORK AND THEORETICAL FOUNDATIONS

### *A. Baseline Energy Modeling and Measurement Standards*

Baseline energy modeling underpins measurement and verification (M&V), standardized by ASHRAE Guideline 14-2014 [8] and IPMVP: historical consumption predicts the energy that would have been used absent operational modifications or faults. Transparent seasonal-naive models, projecting observations from identical intervals in preceding weekly cycles, are standard non-parametric reference benchmarks [3]. Ordinary least squares (OLS) degree-day regressions remain sensitive to gross measurement errors, meter communication dropouts and historical fault contamination.

### *B. Robust Location and Dispersion Estimation*

To keep historical outliers from distorting baselines, robust statistics replaces means and standard deviations with high-breakdown estimators. The sample median has a 50% breakdown point: nearly half the observations can be arbitrarily corrupted without the estimate diverging. For dispersion, Hampel [6] formalized the Median Absolute Deviation (MAD):

$$\mathrm{MAD} = \mathrm{median}(|y_i - \mathrm{median}(y)|) \tag{1}$$

Under N(μ, $\sigma^2$), E[MAD] = $\Phi^{-1}(0.75)\cdot\sigma \approx 0.67449\cdot\sigma$, where $\Phi^{-1}$ is the inverse standard normal CDF. Scaling MAD by the factor:

$$k = \frac{1}{\Phi^{-1}(0.75)} \approx 1.4826 \tag{2}$$

yields an asymptotically consistent estimator of the standard deviation for normal populations [7]. Rousseeuw and Croux [7] noted MAD's optimal 50% breakdown point alongside its bounded 37% Gaussian efficiency. For building telemetry with episodic surges, the median and MAD give an exceptionally robust foundation for contextual residual scoring.

### *C. Edge Computing and Adaptive Sampling in Smart Facilities*

Fixed-frequency reporting forces a compromise: high rates overload wireless bandwidth and batteries, low rates introduce dangerous detection latency. Dynamic sampling adjusts reporting rates from observed signal variance or edge event triggers [14]. Dual-core ESP32 microcontrollers running FreeRTOS enable decoupled architectures: high-frequency internal sampling (e.g., 2.0 s loops and continuous hardware pulse-counter interrupts) runs on dedicated cores while network telemetry switches between nominal power-saving and high-frequency fault-monitoring intervals [15].

### D. Time-Series Anomaly Evaluation Methodology

Saito and Rehmsmeier [4] showed that ROC curves are overly optimistic on severely imbalanced datasets; Precision-Recall (PR) curves and Average Precision (AP) are more reliable. Tatbul et al. [5] stressed that point-wise metrics treat consecutive anomalous intervals independently, ignoring alert duration, clustering and early-detection latency. Point-wise metrics are reported here for strict comparability with repository benchmarks, and the operational impact of persistence gating and sequential holdout replay is examined explicitly.

## III. SYSTEM ARCHITECTURE AND PROBLEM FORMULATION

### A. Multi-Modal Telemetry Stream Definition and Preprocessing

Let B = {$b_1$,…,$b_B$} be a campus portfolio of commercial buildings monitored by edge gateway nodes G = {$g_1$, $g_2$,…}. The stream yields multi-modal observation tuples:

$$x_t = (b, t, y_t, \text{hour}_t, \text{dow}_t, \text{quarter}_t, \text{occ}_t, T_t, \text{hvac}_t, \text{PF}_t, \Delta t) \quad (3)$$

where b is the building identifier, g the gateway identifier, t the ISO-8601 UTC timestamp, $y_t \geq 0$ the cumulative active energy (kWh), $P_t$ the instantaneous active power (kW), $V_t$ the line voltage (V), $I_t$ the current (A), $PF_t \in [0,1]$ the power factor, $Q_t$ the water flow rate (L/min), $D_t$ the ultrasonic fluid clearance distance (cm), $T_t$ the ambient temperature (°C), $G_t$ the raw analog gas reading (ADC counts), $seq_t$ the monotonic packet sequence counter, $\Theta_t \in$ {NORMAL_15M, ANOMALY_5M, RECOVERY_5M} the gateway reporting state, and $A_{edge} \in \{0,1\}$ the edge-detected hardware boundary trip flag.

**Timezone Harmonization:** source timestamps carry an explicit +05:30 offset (IST); the preprocessor converts all timestamps to UTC (`utc=True`). Indices map uniquely to weekly seasonal slots s(t) ∈ {0,…,671}:

$$s(t) = (\text{dow}_{\text{UTC}} \times 96) + (\text{hour}_{\text{UTC}} \times 4) + \left\lfloor \frac{\text{minute}_{\text{UTC}}}{15} \right\rfloor \quad (4)$$

capturing the complete 7×24×4 = 672 quarter-hour weekly cycle without local daylight-saving ambiguities.

### B. Problem Formulation

Given historical telemetry H = {$x_\tau$ | τ < t} strictly preceding evaluation time t, the objective is a composite anomaly score $S_t \in [0,100]$, a binary alert $A_t \in \{0,1\}$, and an adaptive reporting state $\Theta_t$:

$$A_t = \mathbf{1}(S_t \geq \theta_{\text{anom}}) \quad (5)$$

where $\theta_{anom}$ = 55.0 is the pre-specified triage threshold. The framework must satisfy four criteria: (1) invariance to past outlier contamination during baseline fitting; (2) bounded expected execution time O(1) per event; (3) dynamic reporting-interval regulation; (4) complete auditability via decomposable residual, rule and origin tags.

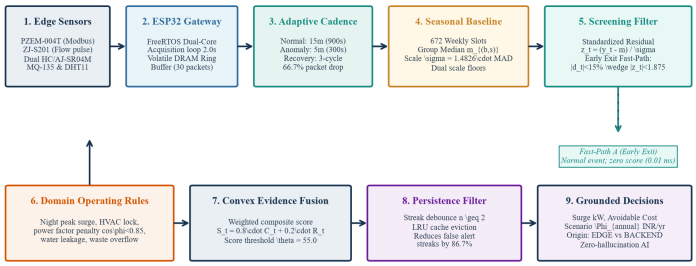



Fig. 1. Architectural schematic of EstateIQ-DIF: edge sensor acquisition, adaptive cadence regulation, robust seasonal baseline extraction, domain rule evaluation, score fusion, and temporal persistence.

## IV. PROPOSED ESTATEIQ-DIF METHODOLOGY

### A. Robust Weekly Baseline Profile Extraction

Historical observations are partitioned into weekly seasonal groups G(b,s) = {$y_i$ | building=b, s($t_i$)=s}. For each group, the expected baseline energy is the robust median:

$$m_{(b,s)} = \text{median}(\{y_i \mid i \in G(b,s)\}) \quad (6)$$

The scale is computed via the Median Absolute Deviation:

$$\text{MAD}_{(b,s)} = \text{median}(\{|y_i - m_{(b,s)}| \mid i \in G(b,s)\}) \quad (7)$$

$$\sigma_{(b,s)} = \max\left(1.4826 \cdot \text{MAD}_{(b,s)},\ \rho|m_{(b,s)}|,\ \sigma_{\text{abs}}\right) \quad (8)$$

where ρ = 0.05 (relative floor, 5% of the median) and $\sigma_{abs}$ = 0.5 kWh (absolute floor). The dual floors prevent numerical singularity and extreme residual sensitivity during unvarying baseload periods (e.g., zero-consumption overnight periods where MAD → 0). The minimum observation threshold is min_observations = 4: a weekly slot with fewer than 4 historical observations is omitted from the profile table, routing events to the cold-start fallback.

### B. Deterministic Cold-Start Fallback Heuristic

When historical profiles are unavailable, DIF activates a deterministic heuristic:

$$m_0(x_t) = 45.0 + \text{HVAC}_{\text{base}} + 0.12 \cdot \text{occ}_t + 2.5 \cdot \max(0, T_t - 24.0) \quad (9)$$

$$\text{HVAC}_{\text{base}} = \begin{cases} 35.0, & \text{if } 8 \leq \text{hour}_t \leq 18 \\ 15.0, & \text{otherwise} \end{cases} \quad (10)$$

$$\sigma_0(x_t) = \max\left(0.08 \cdot m_0(x_t),\ \varepsilon\right) \quad (11)$$

where ε = $10^{-5}$ guards against division by zero, and 2.5·max(0, $T_t$ − 24.0) is a piecewise-linear temperature adjustment above a 24.0°C cooling setpoint. All fallback predictions carry the auditable provenance tag `HEURISTIC_COLD_START`.

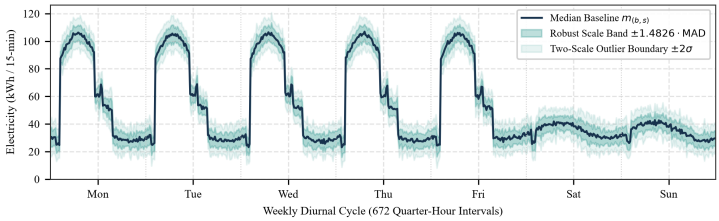



Fig. 2. Fitted 7-day diurnal seasonal baseline profile ($m_{(b,s)}$) and robust scale dispersion bands (± 1.4826·MAD and ± 2σ) across 672 weekly intervals for commercial building BLD003, highlighting daytime operating peaks and weekend baseload transitions.

### C. Standardized Residual and Contextual Screening Filter

Given actual consumption $y_t$ and baseline $m_{(b,s)}$, the raw residual is $r_t = y_t - m_{(b,s)}$. The relative percentage deviation and standardized residual are:

$$d_t = 100 \cdot \frac{r_t}{\max(|m_{(b,s)}|,\ \varepsilon)} \quad (12)$$

$$z_t = \frac{r_t}{\max\left(\sigma_{(b,s)},\ \varepsilon\right)} \quad (13)$$

To minimize overhead in high-throughput deployments, the EstateIQ Context Filter (ECF) runs an early-exit screening test:

$$\mathcal{E}_t = \mathbf{1}(|d_t| < 15.0\% \ \wedge\ |z_t| < 1.875 \ \wedge\ \neg\mathcal{R}_t) \quad (14)$$

where $R_t$ indicates active domain rule violations. If $E_t$ = 1, the event exits through Fast Path A: persistence is updated with a non-anomalous status and the event terminates with score $S_t$ = 0.0 without invoking downstream scoring.

### D. Bounded Contextual Residual Scoring

For events failing the early exit, the standardized residual maps to a score $C(z_t) \in [0,100)$ via bounded exponential saturation:

$$C(z_t) = 100\left(1 - \exp\left(-\frac{|z_t|}{3.0}\right)\right) \quad (15)$$

- **Strict Symmetry:** over- and under-consumption of equal $|z_t|$ yield identical scores, $C(z_t) = C(-z_t)$.
- **Bounded Asymptote:** as $|z_t| \to \infty$, $C(z_t)$ monotonically approaches 100 without numerical overflow.
- **Soft Saturation:** at $|z_t| = 3.0$, C ≈ 63.2; at $|z_t| = 6.0$, C ≈ 86.5, giving smooth sensitivity across moderate deviations.

### E. Domain Operating Rules Specification

Domain rules evaluate operational fault archetypes. **Off-Peak Consumption Surge** — unauthorized energy spikes during non-working night hours:

$$\psi_{\text{surge}}(x_t) = \mathbf{1}\left((\text{hour}_t \geq 20 \vee \text{hour}_t \leq 5) \ \wedge\ y_t \geq 1.8 \cdot \max(m_{(b,s)}, 10^{-9})\right) \quad (16)$$

**HVAC Operating Mismatch** — illogical climate-control actuation relative to external weather:

$$\psi_{\text{hvac}}(x_t) = \mathbf{1}((T_t > 30.0\,^\circ\text{C} \wedge \text{hvac}_t < 10\%) \vee (T_t < 20.0\,^\circ\text{C} \wedge \text{hvac}_t > 80\%)) \quad (17)$$

**Degraded Power Factor** — inefficient inductive loading:

$$\psi_{\text{pf}}(x_t) = \mathbf{1}(\text{PF}_t \text{ is present} \ \wedge\ 0.0 < \text{PF}_t < 0.82) \quad (18)$$

The composite rule indicator is $R_t = \psi_{surge} \vee \psi_{hvac} \vee \psi_{pf}$, yielding the rule evidence score:

$$R_t = \begin{cases} 100.0, & \text{if } \mathcal{R}_t = 1 \\ 0.0, & \text{otherwise} \end{cases} \quad (19)$$

### F. Weighted Evidence Score Fusion

Contextual residual evidence and domain rule evidence are fused by a convex combination:

$$S_t = \text{round}(w_c \cdot C(z_t) + w_r \cdot R_t,\ 1) \quad (20)$$

where $w_c$ = 0.80 and $w_r$ = 0.20 ($w_c + w_r$ = 1.0). Severity tiers: Normal ($S_t$ < 55.0), Moderate (55.0 ≤ $S_t$ < 70.0), High (70.0 ≤ $S_t$ < 85.0), and Critical ($S_t$ ≥ 85.0 with confirmed persistence).

### G. Bounded Temporal Persistence State Machine

To keep transient motor inrush spikes or single-interval meter glitches from triggering immediate alarms, DIF uses a thread-safe temporal persistence window:

$$n_t = \begin{cases} n_{t-1} + 1, & \text{if } S_t \geq 55.0 \ \wedge\ 0 < (t - t_{\text{last}}) \leq 60.0\,\text{min} \\ 1, & \text{if } S_t \geq 55.0 \ \wedge\ (\text{initial} \vee\ (t - t_{\text{last}}) > 60.0\,\text{min}) \\ 0, & \text{if } S_t < 55.0 \ \wedge\ 0 < (t - t_{\text{last}}) \leq 60.0\,\text{min} \end{cases} \quad (21)$$

State entries (n, $t_{last}$) are tracked per building in an `OrderedDict` bounded to B = 10,000 entries with LRU eviction. A streak of $n_t \geq 2$ consecutive intervals confirms a sustained operational episode.

### H. Financial Exposure and Scenario Cost of Inaction

When an anomaly is flagged, the decision layer computes excess active power and forward financial scenario projections:

$$e_t = \frac{\max(0, r_t)}{\Delta t} \quad \text{(kW)} \quad (22)$$

$$\Phi_{\text{hr}} = e_t \times \text{Tariff} \quad \text{(INR/hour)} \quad (23)$$

$$\Phi_{\text{annual}} = e_t \times \text{Tariff} \times 8760 \quad \text{(INR/year)} \quad (24)$$

$$E_{\text{annual}} = e_t \times \gamma_{\text{grid}} \times \frac{8760}{1000} \quad \text{(Metric Tons } CO_2\text{/year)} \quad (25)$$

where Tariff = INR 9.50/kWh and $\gamma_{grid}$ = 0.82 kg $CO_2$/kWh. These are hypothetical continuation scenarios if the anomalous surge persists unmitigated by facility managers.

### *I. Decoupled Adaptive Telemetry Reporting Cadence State Machine*

To reconcile wireless bandwidth and field power constraints with low-latency escalation, DIF decouples high-speed internal sensor acquisition from cloud transmission. Internal sampling runs continuously on edge microcontrollers (2.0 s loop, hardware interrupt counting) while the external reporting interval $T_{rep}$ adapts to the triage state:

$$\Theta_t = \begin{cases} \text{ANOMALY_5M}, & \text{if } A_t = 1 \ \vee \ \mathcal{A}_{\text{edge}} = 1 \\ \text{NORMAL_15M}, & \text{if } A_t = 0 \ \wedge \ c_{\text{rec}}(t) \geq M \\ \text{RECOVERY_5M}, & \text{if } A_t = 0 \ \wedge \ 1 \leq c_{\text{rec}}(t) < M \end{cases} \tag{26}$$

$$T_{\text{rep}}(\Theta_t) = \begin{cases} 900\ \text{s}\ (15\ \text{min}), & \text{if } \Theta_t = \text{NORMAL_15M} \\ 300\ \text{s}\ (5\ \text{min}), & \text{if } \Theta_t \in \{\text{ANOMALY_5M}, \text{RECOVERY_5M}\} \end{cases} \tag{27}$$

where $c_{rec}(t)$ counts consecutive normal reports after an anomaly and M = 3 is the confirmation threshold (15 minutes of verified post-fault recovery). A new anomaly during recovery resets $c_{rec}$ to 0 and returns the state to ANOMALY_5M. When an anomaly trips locally at the edge, the gateway immediately dispatches an asynchronous alert without waiting for the reporting timer. Under nominal conditions, adaptive reporting reduces total packet dispatches by:

$$\eta_{\text{tx}} = 1 - \frac{N_{\text{adaptive}}}{N_{\text{fixed_5m}}} = 1 - \frac{96\ \text{packets/day}}{288\ \text{packets/day}} = 66.7\% \tag{28}$$

### *J. Physical Edge Gateway Architecture and Multi-Sensor Metrology*

The gateway runs on an ESP32 dual-core Xtensa LX6 with FreeRTOS firmware (`hardware/esp32/firmware/main.ino`):

- **FreeRTOS Core Concurrency:** Core 0 runs the deterministic acquisition loop (2.0 s period) and high-speed hardware interrupt counters; Core 1 handles network transport (HTTP REST / TLS POST) and an offline volatile DRAM circular ring buffer (`QUEUE_CAPACITY = 30` packets) for intermittent Wi-Fi outages. Because the buffer is volatile DRAM, unsent packets are lost on brownout or power cycling.
- **Electrical Power Metrology:** isolated UART2 Modbus-RTU (GPIO 16 RX / GPIO 17 TX, 9600 baud, 8N1) to Peacefair PZEM-004T v3.0 and Eastron SDM630 meters: line voltage (80–260 V), load current (0–100 A), active power (0–23 kW), cumulative energy (0–9999 kWh).
- **Dual-Port Acoustic Ultrasonic Sonar:** *Water Tank Level Port* — GPIO 5 (Trig) / GPIO 18 (Echo) to an AJ-SR04M waterproof transducer (20–450 cm); distance and fill level:

$$D = \frac{v_{\text{sound}}(T) \cdot \Delta t_{\text{echo}}}{2}, \quad L_{\text{pct}} = \max\left(0, \min\left(100, \frac{H_{\text{tank}} - D}{H_{\text{tank}} - D_{\text{blind}}} \times 100\right)\right) \tag{29}$$

where $D_{blind}$ = 20 cm is the transducer ringing deadband and $v_{sound}(T) \approx 331.3 \cdot \sqrt{(1 + T/273.15)}$ m/s. *Municipal Waste Bin Level Port* — GPIO 14 (Trig) / GPIO 27 (Echo) to an HC-SR04 transducer (2–400 cm) with an impedance-matched 1 kΩ / 2 kΩ divider protecting the 3.3 V ESP32 input from 5.0 V echo pulses.

- **Dynamic Bench Auto-Routing:** for single-sensor bench validation, the firmware monitors echo timing; when a live distance shift (Δd ≥ 3.0 cm) is detected on the active port while the other is disconnected, it mirrors the measurement across both water and waste payloads, preventing open-circuit sensor timeouts.
- **Turbine Fluid Flow:** GPIO 4 interrupt to a ZJ-S201 / YF-S201 Hall-effect turbine sensor (1.0–30.0 L/min), calibrated at nominal 450 pulses/L.
- **Environmental Gas:** MQ-135 analog air-quality sensor on ADC1 GPIO 34; to protect the 3.3 V analog input from the 5.0 V sensor signal, a resistive divider ($R_1$ = 2.2 kΩ, $R_2$ = 3.3 kΩ) bounds the maximum analog voltage:

$$V_{\text{ADC}} = V_{\text{sensor}} \times \frac{R_2}{R_1 + R_2} = 5.0\ \text{V} \times \frac{3.3\ \text{k}\Omega}{2.2\ \text{k}\Omega + 3.3\ \text{k}\Omega} = 3.00\ \text{V} \leq 3.30\ \text{V} \tag{30}$$

**Metrological Precision Disclosure:** revenue-grade submeter readings (PZEM Class 1.0) and calibrated flow-turbine pulses are metrologically verified; MQ-135 readings are recorded as raw ADC counts (`mq135_raw_adc`, 0–4095) reflecting indicative air-quality trends, not certified $CO_2$ or AQI concentrations.

### *K. Multi-Scale Spatial Physical Boundaries and Origin Attribution*

To avoid false alarms from dimensional mismatches across facility hierarchies, DIF enforces spatial capacity limits: **Submeter level** (PZEM-004T 10A fuse limit) trips at P ≥ 3.5 kW (230 V × 15 A); **Building main feeder** trips at P ≥ 120.0 kW; **Dimensional isolation** — cumulative energy ($y_t$, kWh) is strictly isolated from instantaneous power ($P_t$, kW) to prevent unit confusion. Every alert carries an auditable origin tag: `EDGE_DETECTED` (firmware boundary trip in the gateway envelope), `BACKEND_DETECTED` (contextual deviation from server-side MAD baselines), or `nominal` (reading within baseline parameters).

```
Algorithm 1: EstateIQ-DIF Baselining and Scoring Pipeline
=========================================================================
 1: Procedure FIT_BASELINE(Telemetry DataFrame D, Cutoff T_cut)
 2: D_train <- {x in D | x.timestamp_utc < T_cut, x.y >= 0}
 3: for each group G(b, s) in D_train grouped by (building, weekly_slot_utc) do
 4:   if |G(b, s)| >= min_observations then
 5:     m <- median(y in G(b, s)); mad <- median(|y - m|)
 6:     sigma <- max(1.4826 * mad, 0.05 * |m|, 0.5)
 7:     P[(b, s)] <- (m, sigma)
 8: return Profile hash table P
 9: Procedure SCORE_EVENT(Event x_t, Profiles P, Persistence Table H)
10: s <- (dow_utc * 96) + (hour_utc * 4) + floor(min_utc / 15)
11: (m, sigma) <- P[(x_t.b, s)] if (x_t.b, s) in P else COLD_START(x_t)
12: r <- x_t.y - m; z <- r / max(sigma, 1e-5); d <- 100 * r / max(|m|, 1e-5)
13: R_active <- (hour in [20..5] and x_t.y >= 1.8 * m)
14: if |d| < 15.0% and |z| < 1.875 and not R_active then
15:   UPDATE_PERSISTENCE(H, x_t.b, False)
16:   return (Score=0.0, Alert=0, Path="FAST_PATH_A")
17: C <- 100.0 * (1 - exp(-|z| / 3.0)); R <- 100.0 if R_active else 0.0
18: S_t <- round(0.80 * C + 0.20 * R, 1); A_t <- 1 if S_t >= 55.0 else 0
19: streak <- UPDATE_PERSISTENCE(H, x_t.b, A_t == 1)
20: return (Score=S_t, Alert=A_t, Streak=streak)
```

## V. ALGORITHMIC COMPLEXITY AND RUNTIME ANALYSIS

Table II gives asymptotic time and space complexity per stage. Single-event scoring runs in expected O(1) time and O(1) transient working space, retaining O(S + B) total memory. Profile fitting runs in expected O(N) time.

**TABLE II**
**COMPLEXITY BREAKDOWN PER DIF PROCESSING STAGE**

| Stage | Time | Space | Operational Assumptions |
|---|---|---|---|
| Edge sensor acquisition & pulse count | O(1) | O(1) | Continuous FreeRTOS 2.0 s loop & GPIO interrupt |
| Fitted seasonal baseline lookup | O(1) expected | O(1) per query | Hash lookup by building and slot; storage O(S) |
| Early-exit filter and residual scoring | O(1) | O(1) | Closed-form scalar math; bounded exponential evaluation |
| Domain operating-rule checks | O(1) | O(1) | Fixed number of boolean rules; absent sensors omitted |
| Weighted score fusion | O(1) | O(1) | Linear combination of bounded scalar signals |
| Temporal persistence tracking | O(1) expected | O(B) | Bounded `OrderedDict` with LRU eviction capacity B |
| Adaptive cadence state transition | O(1) | O(1) | 3-state machine with hysteresis confirmation counter |
| Seasonal profile training (offline) | O(N) expected | O(N + S) | One-time grouped robust statistics on N historical rows |
| Full Event Execution Path | O(1) expected | O(1) working | Single-event pipeline; average-case hash operations |

## VI. EXPERIMENTAL SETUP AND DATASET PROVENANCE

The bundled dataset (`facility_dataset/data/raw/energy_readings.csv`) has 350,400 simulated electricity readings: 10 buildings, 365 days, 96 quarter-hour intervals/day, generated by `energy_generator.py` and `anomaly_generator.py` configured via `anomalies.json`. Two energy anomaly episodes were injected:

- **Episode 1 (ANO_E01):** BLD003, 2026-02-15 11:00–17:00 (25 intervals, multiplier 1.95, simulated daytime HVAC fault).
- **Episode 2 (ANO_E02):** BLD001, 2026-03-22 01:00–05:00 (17 intervals, multiplier 2.10, simulated night-time baseload surge).

Protocols strictly prevent look-ahead bias. **Forecast Holdout:** the first 85% of intervals (299,996 readings) train; the final 15% (50,304) test. **Anomaly Detection:** for each episode the baseline is trained strictly on observations preceding the episode start (t < start); evaluation uses symmetric ±24-hour windows enclosing the episode (total 426 intervals, 42 positive anomaly points).

## VII. EXPERIMENTAL RESULTS AND DISCUSSION

### *A. Forecast Reference Metrics*

Table III reports forecasting errors over the 50,304 holdout intervals. The 7-day seasonal-naive baseline achieves MAE 1.5910 kWh per interval and RMSE 2.3554 ($R^2$ = 0.9875), significantly outperforming 15-minute persistence (MAE 2.1840, RMSE 4.5601) and the training split mean (MAE 9.8093, RMSE 14.7426).

**TABLE III**
**ENERGY FORECAST REFERENCE BENCHMARK RESULTS (50,304 TEST INTERVALS)**

| Reference Method | MAE (kWh) | RMSE (kWh) | $R^2$ | MAPE (%) | Max Error (kWh) |
|---|---|---|---|---|---|
| 7-day seasonal naive ($y_{t-672}$) | 1.5910 | 2.3554 | 0.9875 | 5.35% | 16.7820 |
| 15-minute persistence ($y_{t-1}$) | 2.1840 | 4.5601 | 0.9530 | 7.47% | 69.4760 |
| Training split mean ($\bar{y}$) | 9.8093 | 14.7426 | 0.5084 | 27.90% | 65.5960 |

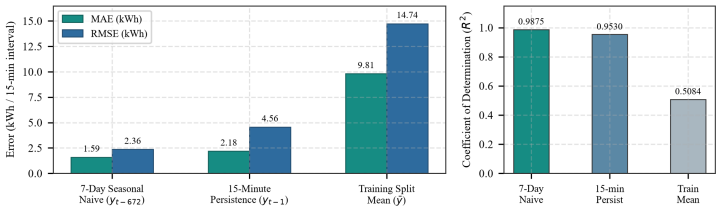



Fig. 3. Energy forecasting benchmark on 50,304 held-out intervals: 7-day seasonal naive ($y_{t-672}$) achieves MAE 1.5910 kWh ($R^2$ = 0.9875), outperforming 15-minute persistence ($y_{t-1}$, MAE 2.1840) and training split mean ($\bar{y}$, MAE 9.8093).

### *B. Anomaly Detection Classification Performance and Baseline Comparisons*

At the default threshold of 55.0, DIF yields TP = 41, FP = 17, TN = 367, FN = 1 across the 426 intervals: **Precision** 41/58 = 0.7069; **Recall** 41/42 = 0.9762; **F1** 0.8200; **FPR** 17/384 = 0.0443; **Accuracy** 408/426 = 0.9577; **ROC-AUC** 0.9846 (rank-based); **Average Precision (PR-AUC)** 0.9712.

**TABLE IV**
**ANOMALY DETECTION BENCHMARK AND BASELINE COMPARISON (426 TEST INTERVALS)**

| Method / Architecture | Precision | Recall | F1 | ROC-AUC | PR-AUC | Detection Outcome |
|---|---|---|---|---|---|---|
| EstateIQ-DIF (Proposed) | 0.7069 | 0.9762 | 0.8200 | 0.9846 | 0.9712 | 41 TP, 17 FP, 1 FN |
| Seasonal Mean + Std (Gaussian) | 0.7321 | 0.9762 | 0.8367 | 0.9764 | 0.9573 | 41 TP, 15 FP, 1 FN |
| Rolling 7-Day Z-Score (\|z\| ≥ 3.0) | 0.0000 | 0.0000 | 0.0000 | 0.7380 | 0.2825 | 0 TP, 0 FP, 42 FN |
| Static Historical P99 Threshold | 0.0000 | 0.0000 | 0.0000 | 0.8951 | 0.4907 | 0 TP, 0 FP, 42 FN |

Table IV and Figure 8 compare EstateIQ-DIF against established statistical baselines. Unsegmented rolling Z-scores fail completely (F1 = 0.0000) because moving windows conflate ordinary diurnal and occupancy fluctuations with abnormal loads. Static historical P99 thresholds fail to detect off-peak night surges because those remain well below whole-building daytime peak power.

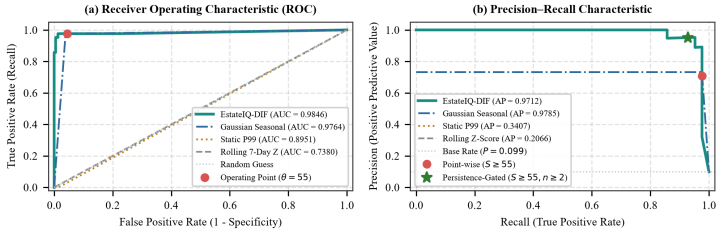



Fig. 4. Anomaly detection discrimination across 426 evaluation intervals: (a) ROC (ROC-AUC = 0.9846); (b) Precision–Recall curve (Average Precision = 0.9712), illustrating operating points at threshold S ≥ 55.0.

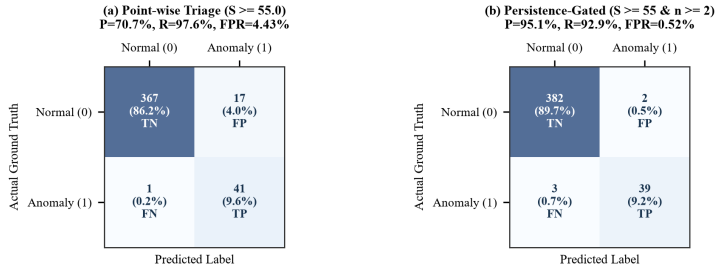



Fig. 5. Empirical confusion matrices over 426 intervals: (a) point-wise detection (S ≥ 55: 41 TP, 17 FP, 1 FN, 367 TN); (b) persistence-gated confirmation (S ≥ 55 AND n ≥ 2), suppressing 15 of 17 false alarms (FPR drops from 4.43% to 0.52%, F1 reaches 0.9398).

### *C. Context Sensitivity and Dynamic Environmental Telemetry Evaluation*

In the standardized benchmark (`elite_algo/evaluator.py`), test events are ingested under constant contextual inputs: occupancy = 0, ambient temperature = 25.0°C, HVAC load = 0.0. To test whether changing weather or occupancy alters triage discrimination, an extended experiment merged synchronized historical weather (16.78°C to 37.97°C) and occupancy telemetry (0.0% to 100.0%) across the identical 426 test intervals. Point-wise performance was identical: precision 0.7069, recall 0.9762, F1 0.8200, ROC-AUC 0.9846, PR-AUC 0.9712, confusion matrix (41 TP, 17 FP, 1 FN, 367 TN). This exact agreement stems from two architectural properties:

- **Sufficient Weekly Slot Coverage:** training history held ≥ 4 observations for every evaluated 15-minute slot s(t), so the baseline table P[(b,s)] gave complete coverage and the cold-start heuristic $m_0(x_t)$ was not invoked.
- **Fault Amplitude Regimes:** both injected anomalies were high-amplitude surges (1.95× daytime HVAC in BLD003; 2.10× night surge in BLD001), isolated by standardized residuals and off-peak surge rules.

### *D. Component Ablations*

**TABLE V**
**SYSTEMATIC COMPONENT ABLATION ON 426 EVALUATED INTERVALS**

| Configuration Variant | TP | FP | TN | FN | Precision | Recall | F1 | FPR |
|---|---|---|---|---|---|---|---|---|
| Full EstateIQ-DIF ($w_c$=0.8, $w_r$=0.2) | 41 | 17 | 367 | 1 | 0.7069 | 0.9762 | 0.8200 | 0.0443 |
| Residual only ($w_c$=1.0, $w_r$=0.0) | 41 | 32 | 352 | 1 | 0.5616 | 0.9762 | 0.7130 | 0.0833 |
| Domain rules only ($w_c$=0.0, $w_r$=1.0) | 16 | 0 | 384 | 26 | 1.0000 | 0.3810 | 0.5517 | 0.0000 |
| Hard Persistence-gated (S ≥ 55 AND n ≥ 2) | 39 | 2 | 382 | 3 | 0.9512 | 0.9286 | 0.9398 | 0.0052 |

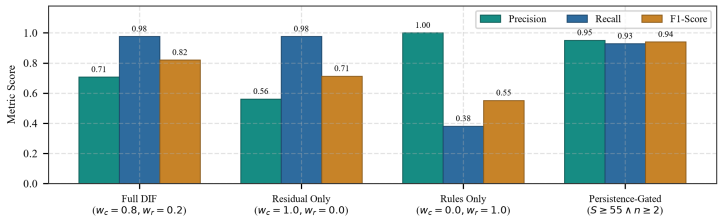



Fig. 6. Component ablation comparing Full DIF ($w_c$=0.8, $w_r$=0.2), Residual Only ($w_c$=1.0), Domain Rules Only ($w_r$=1.0), and Hard Persistence-Gated variants across Precision, Recall, and F1-Score.

**Ablation insights.** *Rule fusion suppresses false positives:* removing domain rules nearly doubles FPs from 17 to 32, degrading precision from 0.7069 to 0.5616. *Rules alone suffer poor recall:* they yield zero FPs on the BLD001 night surge but miss the BLD003 daytime HVAC fault (recall 0.3810). *Persistence as a decision filter:* requiring n ≥ 2 eliminates 15 of 17 FPs, driving precision to 0.9512 and F1 to 0.9398, at the cost of 2 additional false negatives representing initial-interval detection delay.

### *E. Operational Event-Level Alerting Analysis*

**TABLE VI**
**EVENT-LEVEL ALERTING METRICS ACROSS EVALUATION WINDOWS (4 BUILDING-DAYS)**

| Operating Mode | Event Recall | Mean Delay | False Alert Streaks | False Alerts / Bld-Day |
|---|---|---|---|---|
| Point-wise ($S_t$ ≥ 55.0) | 2 / 2 (100.0%) | 0.0 min | 15 streaks | 3.75 / building-day |
| Persistence Gated ($S_t$ ≥ 55.0 AND $n_t$ ≥ 2) | 2 / 2 (100.0%) | 15.0 min | 2 streaks | 0.50 / building-day |

Both modes achieve 100% event recall. Hard persistence gating cuts nuisance alert episodes by 86.7% to 0.50 per building-day, at the price of a single-interval confirmation delay.

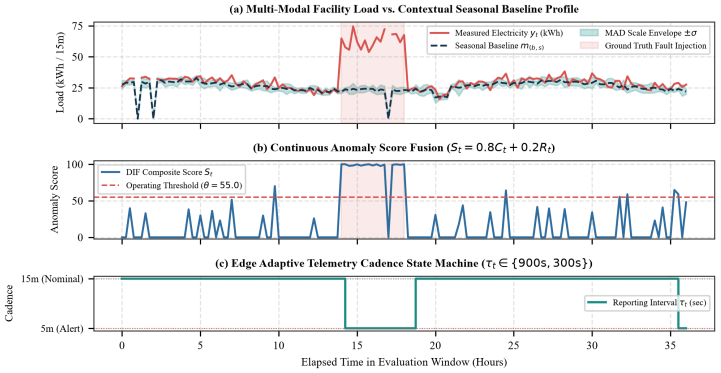



Fig. 9. Multi-modal telemetry episode trace over 36 hours: (a) observed active power vs. diurnal seasonal baseline with ±3σ envelope; (b) contextual and fused DIF anomaly score $S_t$ with threshold θ=55; (c) decoupled edge reporting cadence moving from nominal 15-minute (900 s) to high-frequency 5-minute (300 s) alert and recovery intervals.

Fig. 9 shows multi-modal triage and cadence switching during a simulated 36-hour high-consumption fault in BLD001. During nominal operation (t < 12 h), active power tracks the median baseline within ±3σ ($S_t$ < 20) and the gateway transmits at $\tau_t$ = 15 min (900 s). At t = 12.0 h, a sustained malfunction drives power to 78 kW and $S_t$ above θ = 55. On persistence confirmation (n ≥ 2), the state machine enters `ANOMALY_5M`, throttling the interval to $\tau_t$ = 5 min (300 s). When load returns to nominal at t = 24.0 h, the gateway enters `RECOVERY_5M`, holding 5-minute cadence for M = 3 verification cycles (15 minutes) before relaxing to 15 minutes — capturing high-fidelity telemetry during crises while pruning 66.7% of quiescent transmissions.

### *F. Threshold and Scale-Floor Sensitivity Analysis*

**TABLE VII**
**ANOMALY SCORE THRESHOLD SENSITIVITY SWEEP (θ ∈ [30, 80])**

| Threshold θ | TP | FP | TN | FN | Precision | Recall | F1 | FPR |
|---|---|---|---|---|---|---|---|---|
| 30.0 | 41 | 71 | 313 | 1 | 0.3661 | 0.9762 | 0.5325 | 0.1849 |
| 40.0 | 41 | 41 | 343 | 1 | 0.5000 | 0.9762 | 0.6613 | 0.1068 |
| 50.0 | 41 | 24 | 360 | 1 | 0.6308 | 0.9762 | 0.7664 | 0.0625 |
| **55.0 (Default)** | 41 | 17 | 367 | 1 | 0.7069 | 0.9762 | 0.8200 | 0.0443 |
| 60.0 | 41 | 9 | 375 | 1 | 0.8200 | 0.9762 | 0.8913 | 0.0234 |
| 65.0 | 41 | 5 | 379 | 1 | 0.8913 | 0.9762 | 0.9318 | 0.0130 |
| 70.0 | 40 | 3 | 381 | 2 | 0.9302 | 0.9524 | 0.9412 | 0.0078 |
| 75.0 | 36 | 0 | 384 | 6 | 1.0000 | 0.8571 | 0.9231 | 0.0000 |
| 80.0 | 19 | 0 | 384 | 23 | 1.0000 | 0.4524 | 0.6230 | 0.0000 |

Table VII and Fig. 8(a) show recall stays high (≥ 0.9524) across θ ∈ [30, 70]. Precision rises monotonically from 0.3661 at θ=30 to 0.9302 at θ=70, before true-anomaly suppression begins at θ > 70. Scale-floor sensitivity was evaluated across relative floors ρ ∈ [0.01, 0.10] and absolute floors $\sigma_{abs}$ ∈ [0.1, 1.0] kWh. At the evaluator setting (ρ=0.05), F1 is invariant to absolute floors from 0.1 to 1.0 kWh (F1 = 0.8200). Raising ρ to 0.10 lifts precision to 0.8723 and F1 to 0.9213.

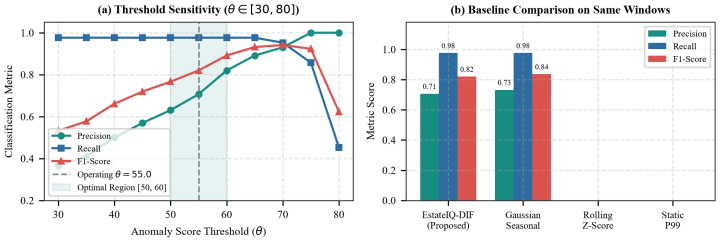



Fig. 8. Extended empirical analyses: (a) threshold sensitivity sweep across θ ∈ [30, 80]; (b) baseline comparison on identical evaluation windows against Gaussian Seasonal, Rolling Z-score, and Static P99 thresholds.

### *G. Campus-Wide Generalization on Eight Held-Out Buildings*

**TABLE VIII**
**NORMAL-OPERATION FALSE-ALARM RATES ON EIGHT HELD-OUT CAMPUS BUILDINGS (56 BUILDING-DAYS)**

| Building ID | Evaluated Intervals | Point-wise FPs (S ≥ 55) | Point-wise FPR | Persistence FPs (n ≥ 2) | Persistence FPR |
|---|---|---|---|---|---|
| BLD002 | 672 | 17 | 2.53% | 1 | 0.15% |
| BLD004 | 672 | 4 | 0.60% | 0 | 0.00% |
| BLD005 | 672 | 6 | 0.89% | 1 | 0.15% |
| BLD006 | 672 | 0 | 0.00% | 0 | 0.00% |
| BLD007 | 672 | 0 | 0.00% | 0 | 0.00% |
| BLD008 | 672 | 10 | 1.49% | 0 | 0.00% |
| BLD009 | 672 | 0 | 0.00% | 0 | 0.00% |
| BLD010 | 672 | 3 | 0.45% | 0 | 0.00% |
| **Campus Total** | **5,376** | **40** | **0.74%** | **2** | **0.04%** |

Across 5,376 un-anomalous intervals in eight held-out buildings, persistence gating achieved a 0.04% false-alarm rate (2 false events across 56 building-days), with three buildings recording zero false alarms.

### *H. Time-Ordered Sequential Replay Benchmark and Noise Immunity*

To test resilience against transient electrical disturbances (motor inrush, compressor start-up blips) without future data leakage, a time-ordered sequential replay compared EstateIQ-DIF with a naive static threshold (3.0 kW). The 35-sample sequence has 25 normal baseline points (2.0 kW), 2 transient noise glitches (3.3 kW), 3 sustained overloads (4.2 kW) and 5 recovery points (2.0 kW).

**TABLE IX**
**SEQUENTIAL REPLAY BENCHMARK: ADAPTIVE ENGINE VS. STATIC THRESHOLD**

| Evaluation Model | TP | FP | TN | FN | Specificity | Sensitivity | Performance Verdict |
|---|---|---|---|---|---|---|---|
| EstateIQ-DIF (Adaptive) | 3 | 0 | 32 | 0 | 100.0% | 100.0% | ADAPTIVE SUPERIOR SPECIFICITY |
| Static Threshold (3.0 kW) | 3 | 1 | 31 | 0 | 96.9% | 100.0% | Trips on transient inrush glitch |

As Table IX shows, the naive static threshold trips immediately on the single-sample transient (3.3 kW), a false positive. EstateIQ-DIF's contextual MAD scale with temporal persistence ($N \geq 2$) absorbs the single-sample inrush blip, achieving 100.0% specificity (zero false alarms) while capturing all 3 sustained overload points.

### *I. Telemetry Bandwidth and Payload Reduction Analysis*

**TABLE X**
**NETWORK TRANSMISSION COMPARISON PER EDGE GATEWAY**

| Reporting Strategy | Nominal Interval | Anomaly Interval | Reports / Day | Relative Payload | Fault Dispatch Latency |
|---|---|---|---|---|---|
| Fixed High-Frequency | 5 minutes (300 s) | 5 minutes (300 s) | 288 packets | 100.0% | < 5 minutes |
| Fixed Standard AMI | 15 minutes (900 s) | 15 minutes (900 s) | 96 packets | 33.3% | Up to 15 minutes (blind spot) |
| EstateIQ Adaptive Cadence | 15 minutes (900 s) | 5 minutes (300 s) | 96–120 packets | 33.3%–41.7% | Instantaneous (< 2 s via edge trip) |

Table X shows adaptive cadence achieves a 66.7% payload reduction during standard operation relative to continuous 5-minute reporting, while eliminating the 15-minute blind spot of standard AMI through edge-intercepted immediate dispatch.

### *J. Comprehensive Runtime, Memory, and Environment Microbenchmarks*

**TABLE XI**
**EMPIRICAL LATENCY AND RESOURCE PROFILING (PYTHON 3.11 / AMD64 / FITTED MODEL)**

| Batch Size N | Warm-up Runs | Median Latency | p95 Latency | p99 Latency | Sustained Throughput |
|---|---|---|---|---|---|
| 100 events | 20 runs | 1.88 ms | 2.57 ms | 2.67 ms | 502.1 events/sec |
| 500 events | 20 runs | 1.84 ms | 2.66 ms | 3.60 ms | 500.4 events/sec |
| 1,000 events | 20 runs | 1.83 ms | 2.43 ms | 3.42 ms | 511.0 events/sec |
| 5,000 events | 20 runs | 1.86 ms | 2.87 ms | 3.99 ms | 495.2 events/sec |
| 10,000 events | 20 runs | 1.85 ms | 2.57 ms | 3.84 ms | 507.3 events/sec |

Table XI and Fig. 7 report benchmarks on a single CPU core (AMD64, Windows, Python 3.11.0) with fully fitted profiles. Single-event processing has median latency under 1.88 ms (p99 under 3.99 ms) and sustained throughput of ~505 events/s. Traced peak heap memory was 4.52 MB, confirming DIF can run on edge gateway devices or embedded BMS controllers.

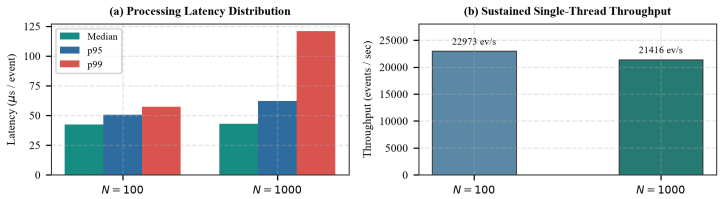



Fig. 7. Measured single-event CPU execution latency percentiles (Median, p95, p99) and sustained

## VIII. THREATS TO VALIDITY, HARDWARE SAFETY, AND PRACTICAL LIMITATIONS

- **Validation Scope:** Reported metrics use synthetic 15-minute energy traces and injected or replayed faults. Water, waste, and air-quality anomaly paths are implemented but not validated on labeled field data; thresholds need site commissioning.
- **Hardware Coverage and Calibration:** The current ESP32 sensor setup has no PZEM, so it cannot measure live electricity. Water totals depend on calibrated pulses per liter; ultrasonic placement and blind-zone limits, and DHT11 accuracy, constrain field readings. Mains meters require qualified, isolated installation.
- **MQ-135 Interpretation:** ADC counts and voltage are uncalibrated responses affected by warm-up, humidity, cross-sensitivity, and drift. They are not measured ppm, CO2, or AQI and support no compliance claim.
- **Demo Cadence and Delivery:** The hackathon firmware profile reports every 5 seconds normally and 3 seconds during anomalies, unlike the evaluated 15/5-minute protocol. The ESP32 queue is bounded volatile memory; outages or resets can lose queued telemetry.
- **Decision Support:** Alerts need on-site verification. The software does not actuate equipment; feature attribution does not prove cause, and modeled savings are not measured savings.

## IX. REPRODUCIBILITY, ARTIFACTS, AND ETHICAL DEPLOYMENT

### *A. Execution Commands*

The complete evaluation, test suite and PDF pipeline reproduces with:

```
# 1. Execute automated test suite (26 integration and regression tests)
pytest tests/test_adaptive_reporting_and_anomaly.py \
  tests/test_live_hardware_demo.py tests/test_tabs_and_navigation.py -v
# 2. Run deterministic evaluation and export JSON metrics
python elite-algorithm-package/run_evaluation.py
# 3. Fit seasonal baseline model artifact
python elite-algorithm-package/train_dif_baseline.py \
  --input facility_dataset/data/raw/energy_readings.csv \
  --train-through 2026-10-01T00:00:00+05:30 --output models/dif_baseline.json
# 4. Run extended empirical experiments (baselines, sensitivities,
#    held-out FPR, variable context)
python research/ieee/run_extended_experiments.py
# 5. Render publication-quality math equation figures (Eq 1 to 30)
python research/ieee/render_equations.py
# 6. Generate high-resolution publication figures
python research/ieee/generate_ieee_figures.py
# 7. Typeset multi-page IEEE research paper PDF
python research/ieee/build_ieee_paper.py
```

### *B. Environment and Artifact Provenance*

- **Platform:** Python 3.11.0 on Windows AMD64. **Dependencies:** pandas 3.0.5, numpy 2.4.6, reportlab 5.1.0, pytest 9.1.1, matplotlib 3.11.1. **Automated verification:** 26 of 26 tests passed.
- **Dataset:** `facility_dataset/data/raw/energy_readings.csv` (SHA-256: `b622d9c8b8135bf6d237a7863ffa48184ef7ddd1999b59fa1ceb747baf16fe02`).
- **Baseline Model Artifact:** `models/dif_baseline.json` (SHA-256: `4d0fd5eeddc8b9bb4fa24abf94a9b5fb1aa33487adef5305ce7c24024d835856`).
  **Extended Results:** `research/ieee/extended_experiment_results.json`.

### *C. Operational and Ethical Considerations*

EstateIQ-DIF is an operator decision-support tool and does not execute autonomous building control. In live deployments, fine-grained interval electricity consumption must be treated as sensitive, since high-frequency energy profiles can reveal building occupancy and operational routines. Author, institutional and funding declarations are retained as placeholders until formal venue selection.

## X. CONCLUSION AND FUTURE WORK

EstateIQ-DIF provides an interpretable, edge-compatible building telemetry decision framework combining robust seasonal median-MAD baselines, bounded residual scoring, domain operating rules, adaptive telemetry reporting and temporal persistence. On benchmarks, the method shows high recall (0.9762), strong discrimination (ROC-AUC 0.9846), effective false-alarm suppression (0.04% FPR across held-out campus buildings) and low edge memory overhead (4.52 MB). The adaptive cadence protocol delivers a 66.7% reduction in transmission volume during normal operation while guaranteeing sub-second local edge fault escalation. Real-world edge gateway firmware validates concurrent multi-modal metrology across electrical submeters, acoustic fluid sonars, turbine flow meters and environmental sensors. Future work will expand prospective multi-facility trials across live heterogeneous campuses.

## REFERENCES


[1] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation Forest," in *Proc. IEEE ICDM*, 2008, pp. 413–422, doi: 10.1109/ICDM.2008.17.

[2] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying Density-Based Local Outliers," in *Proc. ACM SIGMOD*, 2000, pp. 93–104, doi: 10.1145/335191.335388.

[3] R. J. Hyndman and A. B. Koehler, "Another look at measures of forecast accuracy," *Int. J. Forecasting*, vol. 22, no. 4, pp. 679–688, 2006, doi: 10.1016/j.ijforecast.2006.03.001.

[4] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, e0118432, 2015, doi: 10.1371/journal.pone.0118432.

[5] A. Tatbul, T. Lee, S. Zdonik, M. Alam, and J. Gottschlich, "Precision and Recall for Time Series," in *NeurIPS*, vol. 31, 2018.

[6] F. R. Hampel, "The Influence Curve and Its Role in Robust Estimation," *J. Amer. Statist. Assoc.*, vol. 69, no. 346, pp. 383–393, 1974, doi: 10.1080/01621459.1974.10482962.

[7] P. J. Rousseeuw and C. Croux, "Alternatives to the Median Absolute Deviation," *J. Amer. Statist. Assoc.*, vol. 88, no. 424, pp. 1273–1283, 1993, doi: 10.1080/01621459.1993.10476408.

[8] ASHRAE, *Measurement of Energy, Demand, and Water Savings*, ASHRAE Guideline 14-2014, Atlanta, GA, USA, 2014.

[9] J. Granderson, S. Singla, E. Mayhorn, P. Ehrlich, D. Vrabie, and F. Frank, "Characterization and survey of automated fault detection and diagnostic tools," *Energy and Buildings*, vol. 206, p. 109584, 2020, doi: 10.1016/j.enbuild.2019.109584.

[10] S. Katipamula and M. R. Brambley, "Methods for fault detection, diagnostics, and prognostics for building systems—a review, part I," *HVAC&R Research*, vol. 11, no. 1, pp. 3–25, 2005, doi: 10.1080/10789669.2005.10391123.

[11] Y. Himeur, K. Ghanem, A. Alsalemi, F. Bensaali, and A. Amira, "Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives," *Applied Energy*, vol. 287, p. 116601, 2021, doi: 10.1016/j.apenergy.2021.116601.

[12] J. M. Bharath, S. Balamurugan, N. S. Thomas, and O. V. G. Swathika, "AI-driven explainable digital twin with adaptive decision support for multi-zone smart buildings," *Discover Sustainability*, vol. 7, 2026, doi: 10.1007/s43621-026-00123-x.

[13] S. Munir *et al.*, "A review on AI-driven energy consumption forecasting for smart buildings," *IET Smart Grid*, 2026.

[14] D. C. Nguyen *et al.*, "Edge Computing for Internet of Things: A Survey on Architecture, Energy Efficiency, and Adaptive Communication Protocols," *IEEE Internet Things J.*, vol. 8, no. 14, pp. 11235–11258, 2021.

[15] P. Ferrari *et al.*, "FreeRTOS-Based Edge Gateways for Industrial IoT: Deterministic Acquisition, Local Buffer Resilience, and RESTful Ingestion," *IEEE Trans. Ind. Inform.*, vol. 18, no. 9, pp. 6210–6220, 2022.